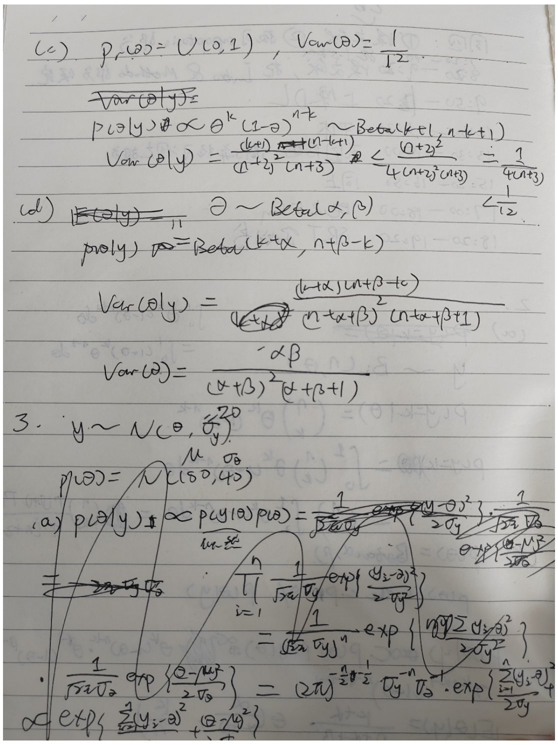


(c) $p_r(\theta) = U(0,1)$, $Var(\theta) = \frac{1}{12}$.

~~Var(θ|y)=~~

$p(\theta|y) \propto \theta^k (1-\theta)^{n-k} \sim Beta(k+1, n-k+1)$

$Var(\theta|y) = \frac{(k+1)(n-k+1)}{(n+2)^2(n+3)} \le \frac{(n+2)^2}{4(n+2)^2(n+3)} = \frac{1}{4(n+3)} < \frac{1}{12}$

(d) ~~E(θ|y) =~~ $\theta \sim Beta(\alpha, \beta)$

$p(\theta|y) = Beta(k+\alpha, n+\beta-k)$

$Var(\theta|y) = \frac{(k+\alpha)(n+\beta-k)}{(n+\alpha+\beta)^2(n+\alpha+\beta+1)}$

$Var(\theta) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

3. $y \sim N(\theta, \overset{20}{\sigma_y})$

$p(\theta) = N(\overset{\mu}{180}, \overset{\sigma_\theta}{40})$

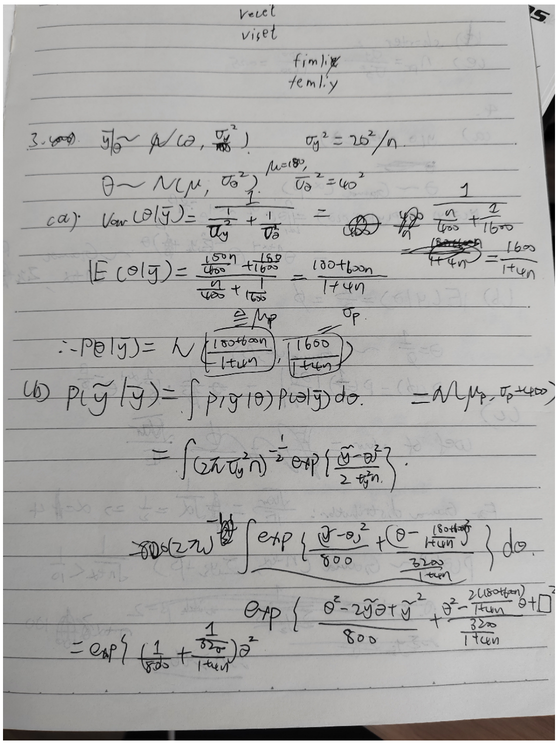
(a) $p(\theta|y) \propto p(y|\theta)p(\theta) = \frac{1}{\sqrt{2\pi}\sigma_y}\exp\{\frac{(y-\theta)^2}{2\sigma_y}\} \cdot \frac{1}{\sqrt{2\pi}\sigma_\theta}\exp\{\frac{(\theta-\mu)^2}{2\sigma_\theta}\}$

$\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma_y}\exp\{\frac{(y_i-\theta)^2}{2\sigma_y^2}\}$

$= \frac{1}{(\sqrt{2\pi}\sigma_y)^n}\exp\{\frac{\sum(y_i-\theta)^2}{2\sigma_y^2}\}$

$\cdot \frac{1}{\sqrt{2\pi}\sigma_\theta}\exp\{\frac{(\theta-\mu)^2}{2\sigma_\theta}\} = (2\pi)^{-\frac{n}{2}-\frac{1}{2}}\sigma_y^{-n}\sigma_\theta^{-1}\cdot\exp\{\frac{\sum_{i=1}^{n}(y_i-\theta)^2}{2\sigma_y}+$

$\propto \exp\{\frac{\sum_{i=1}^{n}(y_i-\theta)^2}{} + \frac{(\theta-\mu)^2}{}\}$



recel
viset

fimliy
temliy

3. $\bar{y}|\theta \sim N(\theta, \frac{\sigma_y^2}{n})$ $\sigma_y^2 = 20^2/n$

$\theta \sim N(\mu, \sigma_\theta^2)$ $\mu = 180$, $\sigma_\theta^2 = 40^2$

(a) $Var(\theta|\bar{y}) = \frac{1}{\frac{1}{\sigma_y^2}+\frac{1}{\sigma_\theta^2}} = \frac{1}{\frac{n}{400}+\frac{1}{1600}} = \frac{1600}{1+4n}$

$E(\theta|\bar{y}) = \frac{\frac{180n}{400}+\frac{180}{1600}}{\frac{n}{400}+\frac{1}{1600}} = \frac{180+600n}{1+4n}$

$\triangleq \mu_p$ σ_p

$\therefore p(\theta|\bar{y}) = N(\boxed{\frac{180+600n}{1+4n}}, \boxed{\frac{1600}{1+4n}})$

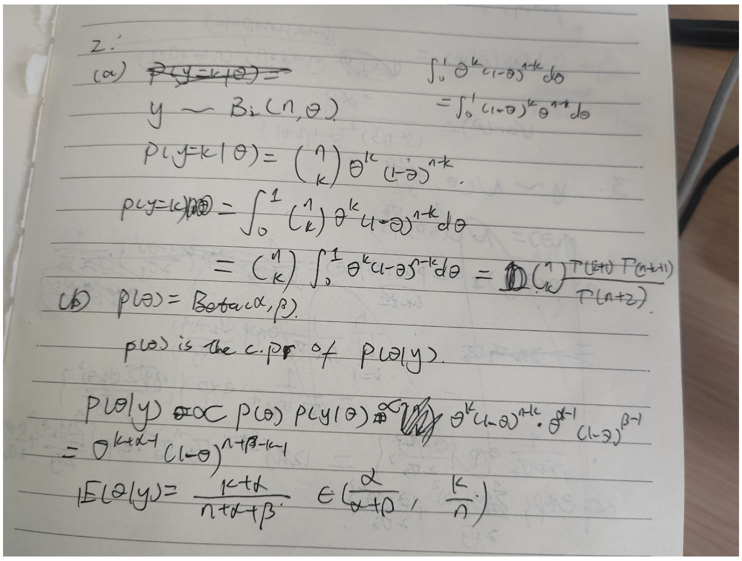
(b) $p(\tilde{y}|\bar{y}) = \int p(\tilde{y}|\theta)p(\theta|\bar{y})d\theta$ $= N(\mu_p, \sigma_p+400)$

$= \int (2\pi\sigma_y^2 n)^{-\frac{1}{2}}\exp\{\frac{(\tilde{y}-\theta)^2}{2\sigma_y^2 n}\}$.

$(2\pi)^{-1}\int \exp\{\frac{(\tilde{y}-\theta)^2}{800} + \frac{(\theta-\frac{180+600n}{1+4n})^2}{\frac{3200}{1+4n}}\}d\theta$.

$\exp\{\frac{\theta^2-2\tilde{y}\theta+\tilde{y}^2}{800} + \frac{\theta^2-\frac{2(180+600n)}{1+4n}\theta+\square^2}{\frac{3200}{1+4n}}$

$= \exp\{(\frac{1}{800}+\frac{1}{\frac{3200}{1+4n}})\theta^2$



2.

(a) ~~P(y=k|θ)=~~ $\int_0^1 \theta^k(1-\theta)^{n-k}d\theta$

$= \int_0^1 (1-\theta)^k\theta^{n-k}d\theta$

$y \sim Bi(n, \theta)$

$p(y=k|\theta) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$.

$p(y=k) = \int_0^1 \binom{n}{k}\theta^k(1-\theta)^{n-k}d\theta$

$= \binom{n}{k}\int_0^1 \theta^k(1-\theta)^{n-k}d\theta = \binom{n}{k}\frac{\Gamma(k+1)\Gamma(n-k+1)}{\Gamma(n+2)}$

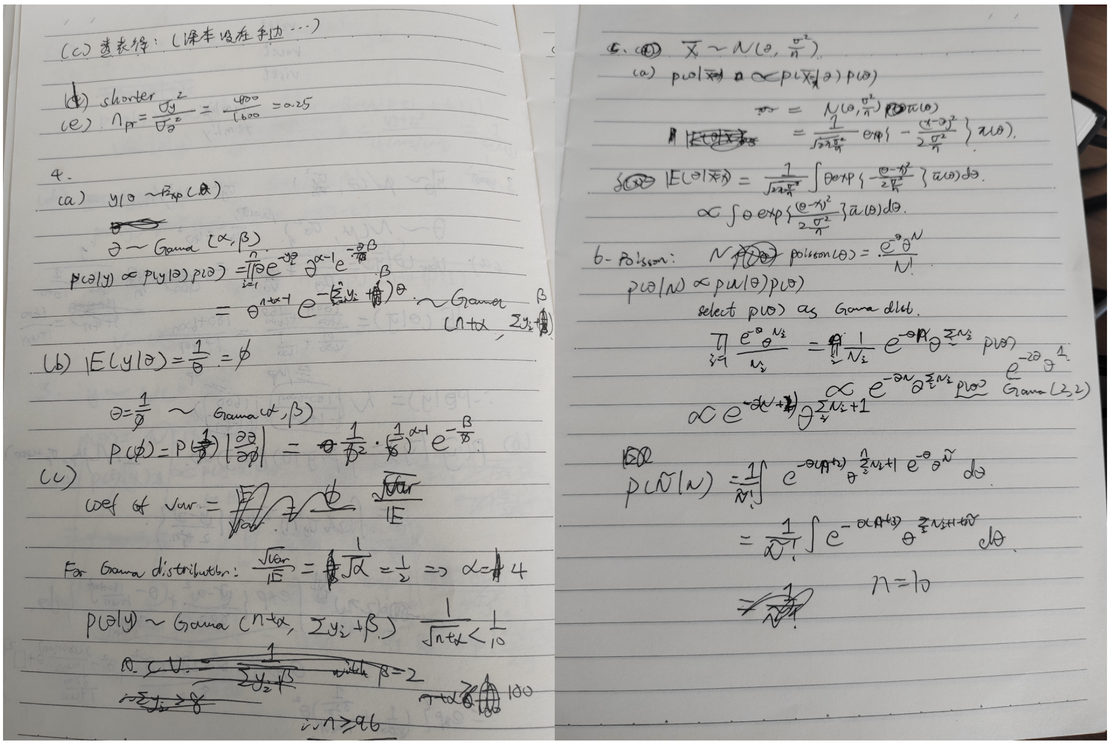
(b) $p(\theta) = Beta(\alpha, \beta)$.

p(θ) is the c.p of p(θ|y).

$p(\theta|y) \propto p(\theta)p(y|\theta) \propto \theta^k(1-\theta)^{n-k}\cdot\theta^{\alpha-1}(1-\theta)^{\beta-1}$

$= \theta^{k+\alpha-1}(1-\theta)^{n+\beta-k-1}$

$E(\theta|y) = \frac{k+\alpha}{n+\alpha+\beta} \in (\frac{\alpha}{\alpha+\beta}, \frac{k}{n})$



(c) 查表得：（课本没在手边···）

(d) shorter

(e) $n_{pr} = \frac{\sigma_y^2}{\sigma_\theta^2} = \frac{400}{1600} = 0.25$

4.

(a) $y|\theta \sim Exp(\theta)$

$\theta \sim Gama(\alpha, \beta)$

$p(\theta|y) \propto p(y|\theta)p(\theta) = \prod_{i=1}^{n} \theta e^{-y_i\theta} \theta^{\alpha-1} e^{-\theta\beta}$

$= \theta^{n+\alpha-1} e^{-(\sum_{i=1}^{n} y_i + \beta)\theta} \sim Gama(n+\alpha, \sum y_i + \beta)$

(b) $\mathbb{E}(y|\theta) = \frac{1}{\theta} = \phi$

$\theta = \frac{1}{\phi} \sim Gama(\alpha, \beta)$

$p(\phi) = p(\frac{1}{\phi})\left|\frac{\partial\theta}{\partial\phi}\right| = \frac{1}{\phi^2}\cdot(\frac{1}{\phi})^{\alpha-1} e^{-\frac{\beta}{\phi}}$

(c)

coef of var $= \frac{\sqrt{Var}}{\mathbb{E}}$

For Gama distribution: $\frac{\sqrt{Var}}{\mathbb{E}} = \frac{1}{\sqrt{\alpha}} = \frac{1}{2} \Rightarrow \alpha = 4$

$p(\theta|y) \sim Gama(n+\alpha, \sum y_i + \beta)$ $\frac{1}{\sqrt{n+\alpha}} < \frac{1}{10}$

~~C.V. $= \frac{1}{\sum y_i + \beta}$ with $\beta = 2$~~

~~$\therefore \sum y_i \geq 8$~~

$n+\alpha \geq 100$

$\therefore n \geq 96$

5. $\bar{X} \sim N(\theta, \frac{\sigma^2}{n})$

(a) $p(\theta|\bar{x}) \propto p(\bar{x}|\theta)p(\theta)$

$= N(\theta, \frac{\sigma^2}{n})\pi(\theta)$

$= \frac{1}{\sqrt{2\pi\frac{\sigma^2}{n}}} \exp\{-\frac{(x-\theta)^2}{2\frac{\sigma^2}{n}}\}\pi(\theta)$

$\mathbb{E}(\theta|\bar{x}) = \frac{1}{\sqrt{2\pi\frac{\sigma^2}{n}}} \int \theta \exp\{-\frac{(\theta-x)^2}{2\frac{\sigma^2}{n}}\}\pi(\theta)d\theta$

$\propto \int \theta \exp\{\frac{(\theta-x)^2}{2\frac{\sigma^2}{n}}\}\pi(\theta)d\theta$

6. Poisson: $N \sim poisson(\theta) = \frac{e^{-\theta}\theta^{N}}{N!}$

$p(\theta|N) \propto p(N|\theta)p(\theta)$

select $p(\theta)$ as Gama dist.

$\prod_{i=1} \frac{e^{-\theta}\theta^{N_i}}{N_i} = \prod_i \frac{1}{N_i} e^{-\theta n} \theta^{\sum N_i} p(\theta)$

$e^{-2\theta}\theta^{1}$ Gama(2,2)

$\propto e^{-\theta n}\theta^{\sum_i N_i} p(\theta)$

$\propto e^{-\theta(n+2)}\theta^{\sum_i N_i + 1}$

$p(\tilde{N}|N) = \frac{1}{\tilde{N}!}\int e^{-\theta(n+2)}\theta^{\sum_i^n N_i + 1} e^{-\theta}\theta^{\tilde{N}} d\theta$

$= \frac{1}{\tilde{N}!}\int e^{-\theta(n+3)}\theta^{\sum_i N_i + 1 + \tilde{N}} d\theta$

$n = 10$